

DropBox

Functional requirements:

user should be able to upload/download any file based on his preference,
user should be able to search for any file among the existing file based on his preference,
user should be able to mark favorites based on his preference,

Non-functional requirements:

Data-partitioning,

load balancing ,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency,

high throughput,

CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime i.e

Availability>>Consistency)

Core Entities:

user,

file,

metadata,

API :

POST:/upload

{

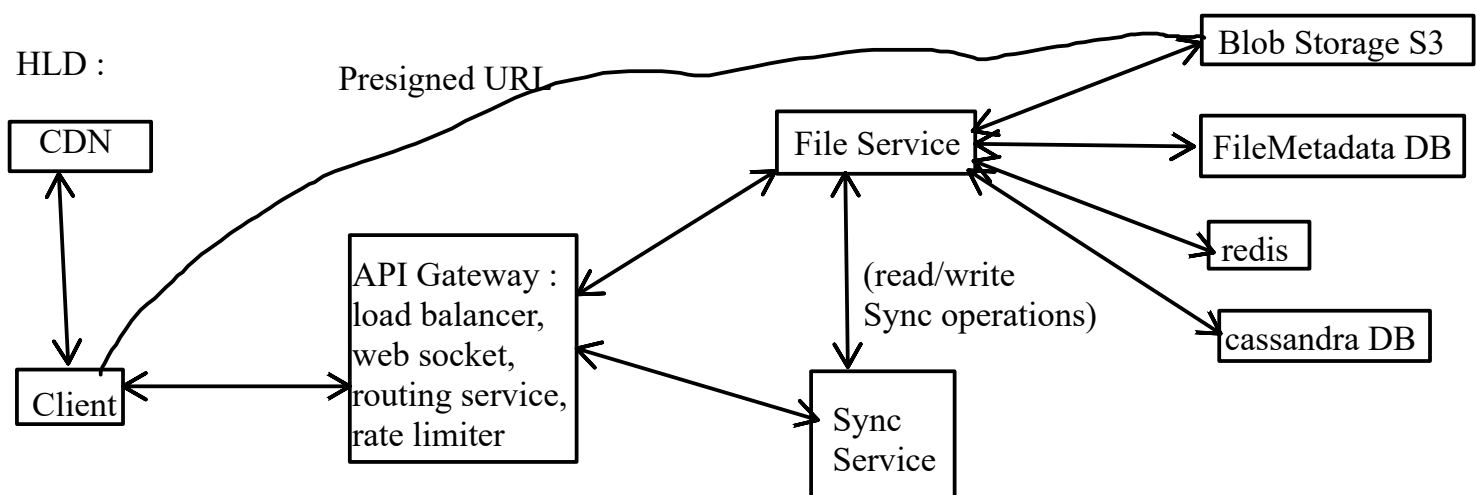
FileID,

FileMetadata

}

GET:/search/{fileID}

HLD :



- 1) we need to chunk 50GB file into multiple pieces
- 2) we need to fingerprint/hash each piece of these files separately
- 3) we need to upload each piece of these files separately
- 4) we need to update completed status for each piece of these chunks separately